Name: ______________________ Score: ________

Directions: Please answer the questions in the space provided. To get full credit you must show all of your work. Use of calculators and other computing or communication devices is **not** allowed on this test.

1. This problem involves the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined as $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 4x-4y+2z \\ -4x+4y-2z \\ 2x-2y+z \end{bmatrix}$.

   (a) Find the preimage of $\mathbf{v} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$

   The preimage is the set of all vectors $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ satisfying $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 4x-4y+2z \\ -4x+4y-2z \\ 2x-2y+z \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$

   Finding these vectors involves solving the system $\left\{\begin{array}{rcr} 4x-4y+2z & = & 2 \\ -4x+4y-2z & = & -2 \\ 2x-2y+z & = & 1 \end{array}\right.$

   $$\begin{bmatrix} 4 & -4 & 2 & 2 \\ -4 & 4 & -2 & -2 \\ 2 & -2 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 1/2 & 1/2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \left\{\begin{array}{l} x = \frac{1}{2}+s-\frac{1}{2}t \\ y = s \\ z = t \end{array}\right.$$

   Thus, the preimage is the set $\boxed{\left\{\begin{bmatrix} \frac{1}{2}+s-\frac{1}{2}t \\ s \\ t \end{bmatrix} : s,t \in \mathbb{R}\right\}}$

   (b) Find the standard matrix for T.

   Since $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 4x-4y+2z \\ -4x+4y-2z \\ 2x-2y+z \end{bmatrix} = \begin{bmatrix} 4 & -4 & 2 \\ -4 & 4 & -2 \\ 2 & -2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$, it follows that the standard matrix is $\boxed{A = \begin{bmatrix} 4 & -4 & 2 \\ -4 & 4 & -2 \\ 2 & -2 & 1 \end{bmatrix}}$

   (c) Find the kernel of T.

   This is the nullspace of A.

   Doing the usual computations, $\begin{bmatrix} 4 & -4 & 2 & 0 \\ -4 & 4 & -2 & 0 \\ 2 & -2 & 1 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \left\{\begin{array}{l} x = s-\frac{1}{2}t \\ y = s \\ z = t \end{array}\right.$ Thus

   $$\boxed{\ker(T) = \left\{\begin{bmatrix} s-\frac{1}{2}t \\ s \\ t \end{bmatrix} : s,t \in \mathbb{R}\right\} = \left\{ s\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1/2 \\ 0 \\ 1 \end{bmatrix} : s,t \in \mathbb{R}\right\} = \operatorname{span}\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 0 \\ 1 \end{bmatrix}\right)}.$$

   (Any of these three forms is acceptable.)

   (d) Find the range of T

   The range of T is the column space of A, that is the span of the columns of A. Looking at A (part b), you can see that each column is a multiple of the first one . Therefore $\boxed{\operatorname{Range}(A) = \operatorname{span}\left(\begin{bmatrix} 4 \\ -4 \\ 2 \end{bmatrix}\right)}$.

2. Suppose $T : P_2 \to P_2$ is a linear transformation for which $T(1) = x$, $T(x) = 1 + x$, and $T(x^2) = 1 + x + x^2$. Find $T(2 - 6x + x^2)$.

$$T(2-6x+x^2) = T(2) - T(6x) + T(x^2) = 2T(1) - 6T(x) + T(x^2) = 2x - 6(1+x) + 1 + x + x^2 = \boxed{-5 - 3x + x^2}$$

3. Suppose $T : \mathbb{R}^2 \to \mathbb{R}^2$ is counterclockwise rotation of $135°$ in $\mathbb{R}^2$. Find the standard matrix for T.

$$A = [T(\mathbf{e}_1)\ T(\mathbf{e}_2)] = \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

4. Suppose $T : P_2 \to P_3$ is defined as $T(p) = xp$.
For example: $T(2 - x^2) = x(2 - x^2) = 2x - x^3$.
Find the matrix for T relative to the bases $B = \{1, x, x^2\}$ and $B' = \{1, x, x^2, x^3\}$.

$$A = [\ [T(1)]_{B'}\ \ [T(x)]_{B'}\ \ [T(x^2)]_{B'}\ ] = [\ [x]_{B'}\ \ [x^2]_{B'}\ \ [x^3]_{B'}\ ] = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Checking, $A[a + bx + cx^2]_B = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ a \\ b \\ c \end{bmatrix} = [ax + bx^2 + cx^3]_{B'} = [T(a + bx + cx^2)]_{B'}$.

5. Determine if the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined as $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x+y \\ x-y \end{bmatrix}$ has an inverse. If it does, find its inverse.

Since $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x+y \\ x-y \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$, it follows that the standard matrix for T is $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, and T is invertible if A is invertible, and if this is the case A^{-1} is the standard matrix for T^{-1}.

Now, $A^{-1} = \dfrac{1}{1\cdot(-1) - 1\cdot 1}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}$.

Thus T is invertible and $T^{-1}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{x+y}{2} \\ \frac{x-y}{2} \end{bmatrix}$

6. Suppose a linear transformation $T : M_{3,3} \to \mathbb{R}^7$ has rank 5. What is the dimension of $\ker(T)$?

We know $\dim(M_{3,3}) = \text{rank}(T) + \text{nullity}(T)$.
Since $\dim(M_{3,3}) = 3 \cdot 3 = 9$, this becomes
$9 = 5 + \text{nullity}(T)$, so $\text{nullity}(T) = 4$.

Recall $\text{nullity}(T) = \dim(\ker(T))$, so $\boxed{\text{the kernel has dimension 4.}}$

7. Consider the following linear transformations.

$T_1 : \mathbb{R}^2 \to \mathbb{R}^2$ is defined as $T_1\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x-2y \\ 2x+3y \end{bmatrix}$

$T_2 : \mathbb{R}^2 \to \mathbb{R}^2$ is defined as $T_2\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x \\ x-y \end{bmatrix}$

Find the standard matrix for $T_1 \circ T_2$.

$T_1\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x-2y \\ 2x+3y \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$, so T_1 has standard matrix $\begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}$.

$T_2\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x \\ x-y \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$, so T_2 has standard matrix $\begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}$.

Thus the standard matrix for $T_1 \circ T_2$ is $\begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 7 & -3 \end{bmatrix}$